foresea

Judge's guide

Foresea predicts where an unknown underwater drone is going and moves only the boats in its way, so the port never has to shut down. On normal days the same system protects divers, whales and crews.

Try it in four minutes

Open the site. Go to `https://foresea.imdusty.com` on a laptop. You get your own private harbor, so nothing you do affects anyone else.

Take the tour. Click **Start the tour**. It runs the real scenes for you, from a live harbor to a night incident to an ordinary Tuesday, and points at each part as it happens.

Be the captain. When the tour reaches "Be the captain", scan the Harbor Launch 7 QR code with your phone camera and tap Join. Your phone gets the alert the moment the officer publishes.

Switch views any time. The **Views** button in the top right moves between the harbor desk, the captain phone, the drive page and the public board.

What to look for

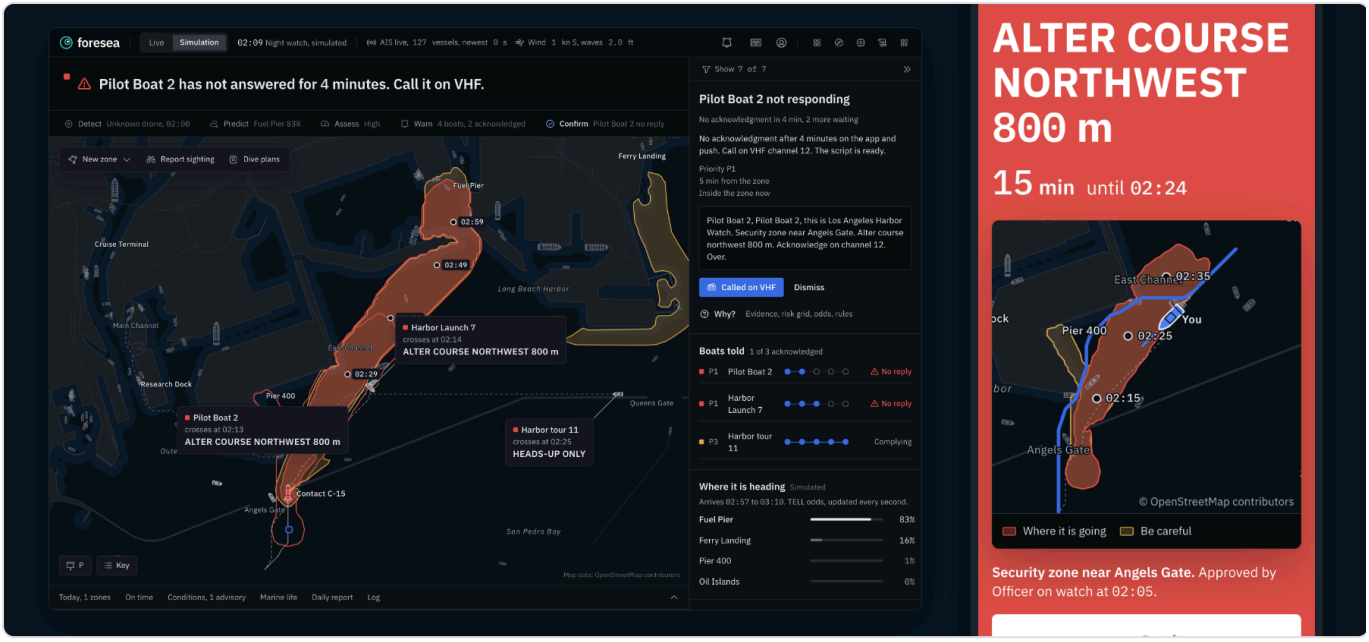
The forecast. Yellow is everywhere the drone could be in the next half hour. Red is where it is most likely going, with time marks. It narrows as the drone commits.

The gentlest notice. Instead of closing the port, Foresea moves only the boats whose routes cross the forecast and tells you the delay in minutes. Nothing closes.

A person decides. The officer publishes every public notice with one tap. High risk always waits for a person, and every decision has a **Why?** button that shows the evidence, the risk grid, the odds and the rules.

Drive it yourself. Open **Views > Drive the drone** on your phone, grab the joystick, and see if the desk calls your target before you get there.

The everyday side. Dive plans, a dredge, whale slow zones from captain reports, live tide and weather on the phone, a lost research robot turned into a search area, and a person overboard button that drops a search area drifting with the real current.



One tap from the officer, and the captain in the path gets one clear instruction, a countdown and a new route.

What is real

Live ship positions from AIS. Tides, currents, wind, waves and water temperature from NOAA stations and buoys. Marine forecasts and advisories from the National Weather Service. The chart from OpenStreetMap. Notice wording and translations from IBM Granite running on our own server.

Simulated, and labeled on screen: the underwater contacts and their sensor readings, and the small boats with phones.

Proof

In 1,000 simulated nights detected at the harbor gates, Foresea gave about **74 minutes** of warning against about 47 for a nearest dock guess, and picked the right target before arrival every time.

The drone stayed inside the yellow forecast **97%** of the time. The traffic model flags about 0.5% of normal real traffic and catches 94% of ships that go dark.

353 automated tests and two full walkthroughs pass on every change.

How it works

TELL, our prediction model, uses Bayesian inverse planning: every move a contact makes raises the odds of the targets it is getting closer to over water, and two extra ideas, "lost" and "drifting with the current", keep a dead robot from looking like an attack. Risk is likelihood times severity on a 5 by 5 grid with five hard rules on top.

Everything is TypeScript. One engine runs in the browser, a background worker and the server. React and MapLibre draw the desk and the phone app on an offline nautical chart. A Node server with Socket.IO keeps everyone in sync and logs every action. Phones get real push alerts and keep working offline.

Responsible use

Foresea never touches any vehicle. A person approves every public notice, any action can be undone with one tap, and everything is logged with who did it and when. Captains never see risk scores.

Foresea, InnovateLA Dual-Use Hackathon, Marine track with NULL Systems.